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Date	**14.10.2026**

COURSE CODE: ECA1009

COURSE NAME: Microprocessor and Micro Controller

Experiment 1

Analysis, Development and Verification of 8-bit and 16-bit Arithmetic Operations Using Assembly Language Program

Sub-Experiments:

A. Develop and Verification of 8-Bit Addition and Subtraction Using Intel 8-Bit Microprocessor Assembly Language Programming

B. Design and Verification of 16-Bit Addition and Subtraction Using Intel 8-Bit Microprocessor Assembly Language Programming

C. Implementation and Testing of 8-Bit Multiplication and Division Using Intel 8-Bit Microprocessor Assembly Language Programming

D. Implementation and Testing of 16-Bit Multiplication and Division Using Intel 8-Bit Microprocessor Assembly Language Programming

LABORATORY PERFORMANCE ASSESSMENT RUBRIC

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weight (%)	Marks
Develop 8-bit and 16-bit addition and subtraction programs	Program Development & Execution	Develops and executes correct programs independently with proper memory addressing and result storage	Correct work with minor errors and limited assistance	Partially correct; needs considerable assistance	Unable to complete or results incorrect	25	
Develop 8-bit and 16-bit multiplication and division programs	Loop & Register-Pair Handling	Correctly uses the counter loop and the BC, DE and HL register pairs and stores product, quotient and remainder at the correct locations	Correct work with minor errors and limited assistance	Partially correct; needs considerable assistance	Unable to complete or results incorrect	25	
Analyze the flag register	Flag Analysis & Interpretation	Predicts and verifies all flags correctly and clearly distinguishes carry from overflow	Correct work with minor errors and limited assistance	Partially correct; needs considerable assistance	Unable to complete or results incorrect	30	
Verify results against theory	Manual Verification & Justification	Verifies all results by manual hex calculation with zero deviation and justifies addressing modes and the zero-divisor condition	Correct work with minor errors and limited assistance	Partially correct; needs considerable assistance	Unable to complete or results incorrect	20	